

Arsh Singh

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TRAINING

Advanced Python for ML & AI | CSE Pathshala| certificate **Jun' 25 - Jul' 25**
Python| NumPy| Pandas| Matplotlib| Scikit-learn

- Completed a 7-week industry-focused training in Advanced Python, Data Science, and Machine Learning With hands-on implementation of end-to-end ML workflows.
- Built ML solutions using data preprocessing, feature engineering, model training, hyperparameter tuning, and performance evaluation on real-world datasets.
- Implemented Regression, Decision Trees, Random Forest, SVM, KNN, Naïve Bayes, and K-Means using Scikit-learn for classification, regression, and clustering tasks.

PROJECTS

AI Based Job Recommendation System **Jan' 26 - Mar' 26**
Python| Django| Sentence Transformers| Scikit-learn

- Developed an NLP-based recommendation system to parse PDF resumes, perform text preprocessing, and extract structured candidate features including skills, education, and experience.
- Built a semantic retrieval pipeline using Sentence Transformers to generate resume embeddings and computed cosine similarity with job embeddings for Top-N recommendation.
- Designed a feature-based ranking pipeline integrating semantic similarity scores, skill matching, experience, salary, and location to improve recommendation relevance and retrieval performance.

Github Link: <https://github.com/Arsh4779/AI-Based-Job-Recommendation-System>

Anomaly Detection In Hyperspectral Image Processing **Apr' 25 - Jun' 25**
Python| Scikit-learn| TensorFlow/Keras| OpenCV

- Created an end-to-end anomaly detection pipeline for high-dimensional hyperspectral images, Incorporating spectral preprocessing, normalization, and PCA-based dimensionality reduction to optimize feature representation.
- Implemented and evaluated unsupervised and one-class learning models, including One-Class SVM, Isolation Forest, and Autoencoders for anomaly localization in hyperspectral data.
- Assessed model performance using Precision, Recall, F1-Score, and anomaly maps, enabling robust detection across high-dimensional spectral datasets with reduced computational complexity.

Github Link: <https://github.com/Arsh4779/Anomaly-Detection-in-Hyperspectral-Image-Processing>

CERTIFICATES

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|---|----------------|
| • Social Networks <i>NPTEL</i> <u>certificate</u> | Aug' 25 |
| • Computer Communications <i>Coursera</i> <u>certificate</u> | Nov' 24 |
| • The Bits and Bytes of Computer Networking <i>Google</i> <u>certificate</u> | Sep' 24 |

SKILLS

Languages:	C++, Java, C, Python, JavaScript, SQL
Frameworks:	Streamlit, TensorFlow, Keras, Django, React, HTML, CSS
Tools/Platforms:	MySQL, GitHub, Jupyter Notebook, Google Colab, VS Code, Linux, Windows
Skills:	Problem-Solving Skills, Adaptability, Time Management, Leadership

EDUCATION

Lovely Professional University <i>Bachelor of Technology - Computer Science and Engineering; CGPA: 7.14</i>	Aug' 23 - Jul' 27 Jalandhar, Punjab
Gajera International School <i>Intermediate; Percentage: 74</i>	Apr' 20 - Mar' 22 Surat, Gujarat
Vigyanand Kendriya Vidyalaya <i>Matriculation; Percentage: 93</i>	Apr' 19 - Mar' 22 Siwan, Bihar